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Non-targeted facilitation of primary sludge anaerobic fermentation by micro-aeration and the simultaneous nutrients transformations

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ABSTRACT

Anaerobic fermentation is a widely used technology for resource recovery from municipal sludge (e.g., primary sludge). Micro-aeration is an emerging strategy which can be incorporated into this technology. However, the effect of micro-aeration on the anaerobic fermentation performance and the nutrients transformations are not well known. This study investigated the impact of micro-aeration on the production of volatile fatty acids (VFAs) from primary sludge as well as the simultaneous nitrogen (N) and phosphorus (P) transformations in the system. The obtained results indicated that application of micro-aeration improved the microbial activities and facilitated all the hydrolysis, acetogenesis and methanogenesis processes. Hydrogenotrophic methanogen was enriched by micro-aeration while the acetoclastic methanogenesis was still the dominant CH_4 production pathway at gene level. The concentration of VFAs was lower with micro-aeration than in the control system, which could be attributed to the enhanced consumption of VFAs. The simultaneous transformations of N and P were observed with different patterns. Micro-aeration led to removal of up to 131.6 mg/L of N, with occurrence of nitrification and denitrification found to be responsible for the N removal. In contrast, the total P content remained unchanged in the reactor. However, micro-aeration was found to inhibit both extracellular organic-P mineralization and intracellular poly-P degradation, resulting in lower soluble PO_4^{3-}-P and more P distributed in the organic-P fraction in solid phase. This study provides novel insights into the microscopic mechanisms of the micro-aeration on anaerobic fermentation and the simultaneous regulation of N and P release.

1. Introduction

Resource recovery from municipal sludge is of great importance considering that treatment and disposal of sludge account for approximately 20–60 % of the operating costs of wastewater treatment plants (WWTPs) [63]. Anaerobic fermentation is widely used in sludge stabilization and resource recovery (e.g., in the form of volatile fatty acid (VFA)). As an important component of WWTP sludge, primary sludge contains a large fraction of readily bioavailable organic matters, which result in efficient VFA production compared to waste activated sludge [30]. The obtained fermentation broth rich in VFA can be used as the carbon source for wastewater treatment or biosynthesis of other value-added commodities [49,52].

In order to facilitate the hydrolysis step and enhance the VFA production during anaerobic fermentation, various research have been carried out on operational parameters optimization (e.g., temperature, sludge retention time (SRT)) as well as application of pretreatment methods (e.g., ultrasound, thermal hydrolysis) [9,37]. For example, Wang et al. found that lower temperatures (20–35 °C) were not conducive to VFA production in two-phase reactors, mainly due to resulting lower microbial and enzyme activities [47]. Miron et al. found that a shorter SRT (<8d) was more favorable for VFA production from primary sludge [33], which was also supported by Yuan et al. [59]. Beside operating parameters optimization, VFA production could also be enhanced by applying physical pretreatment such as ultrasound [19] and thermal hydrolysis [34]. Chemical methods such as acid [6] and alkali pretreatment [38,39] have also been explored to improve VFA production. However, these pretreatment methods often require

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intensive energy input or the addition of extra chemicals, thus leading to increased operating costs and production of chemical wastes [52].

Alternatively, apart from the above-mentioned methods, micro-aeration is an emerging strategy used in anaerobic fermentation, due to its environmental friendliness and economic efficiency [10]. The presence of oxygen has been shown to enhance the activity and diversity of facultative microorganisms which utilizing O_2 as the electron acceptor [24]. Lim et al. found that micro-aeration increased the abundances of phyla Firmicutes and Bacteroidetes to 72 % and 24 % respectively, leading to enhanced hydrolysis efficiency due to extracellular hydrolase secreted by Firmicutes [24]. In addition, reactive oxygen species induced by micro-aeration can also promote the intracellular enzyme activity of microorganisms [60]. An et al. studied the hydrolysis kinetics of industrial park wastewater with limited aeration, and found that the introduction of oxygen increased the reaction rate (*k* parameter) by 80.8 % compared to the control group [2]. It has been reported in most studies to reduce the accumulation of VFA and increase CH_4 production [35]. In contrast, some other studies also reported that introduction of oxygen can inhibit the growth of methanogens, avoiding VFA being consumed by methanogens [52,58]. The primary sludge contains a high portion of organic matter and was reported to be more biodegradable than waste activated sludge [53], but the effects of micro-aeration on its VFA production performance remains unclear. In addition, a large amount of nitrogen and phosphorus would be released during anaerobic fermentation, which can increase the N and P load during its subsequent utilization (e.g., as carbon source for wastewater treatment) [29]. Biological N and P removal during wastewater treatment depend on alternating anaerobic and aerobic conditions, which is commonly realized in separate bioreactors [4,17]. Hence, nitrification and P uptake are not expected to happen during anaerobic fermentation, resulting in high NH_4^+-N and PO_4^{3-}-P accumulation. But micro-aeration could introduce aerobic micro-regions under the anaerobic environment, in which nitrification and P release might happen [21]. Nevertheless, the existing researches primarily focused on the anaerobic fermentation performance while overlooked the simultaneous transformations of soluble N and P [23,58]. Therefore, the impact of micro-aeration on primary sludge fermentation as well as the accompanying N and P transformations needs to be comprehensively investigated to advance this strategy towards better application.

Based on the above information, this study aims to explore the effects of micro-aeration on anaerobic fermentation of primary sludge and the mechanism of N and P transformations. The effects of micro-aeration on organic matters solubilization, N and P transformation and VFA production performance during primary sludge fermentation were characterized using batch and semi-continuous experiment. To this end, the activity variations of key enzymes involved in hydrolysis, acetogenesis and methanogenesis steps were investigated with and without micro-aeration. Finally, metagenomic analysis was used to reveal the mechanism of observed changes at a gene level. Altogether, these findings provide an in-depth understanding of the effect of micro-aeration on anaerobic fermentation systems and the underlying mechanism.

2. Materials and methods

2.1. Source of primary sludge and inoculum

The primary sludge was obtained from the primary sedimentation tank of a full-scale WWTP in Chongqing, China. The obtained primary sludge was screened through an 8-mesh sieve to remove large inorganic substances, and stored at 4 °C before use. The inoculum sludge was obtained from a semi-continuous anaerobic fermentation reactor for VFAs production, which was operated steadily at 37 °C with an SRT of 6 days, and mechanical stirring of 100 rpm. The main characteristics of primary sludge and inoculum are reported in Table S1 (Supporting Information, SI).

2.2. Batch experimental setup

The batch experiments were conducted in 2 sets of serum bottles, which compromised a control, and a micro-aeration group. Each serum bottle (500 mL working volume) was connected to a gas bag and fed with 400 mL of primary sludge. Bottles without aeration were prepared as control group. Micro-aeration was realized using a peristaltic pump connected to the bottle, and operated to supply 0.001 vvm (i.e., 4 mL/min) of air based on previous study [52]. After flushing with N_2 for 5 min, all the bottles were sealed and placed in a water bath shaker at 37 °C and 100 rpm for 9 days. Gas and liquid samples were collected by a syringe every day for gas components and other determinations.

2.3. Operation of semi-continuous reactors

The semi-continuous experiment was conducted in 2 continuous stirred tank reactors (CSTR) (working volume of 4 L) made of plexiglass to study the long-term influence of micro-aeration. Water bath was designed to maintain the temperature around 37 °C during the fermentation. The aeration diffuser was placed at the bottom of the reactor, and micro-aeration was achieved same to the bath experiment. The reactor with micro-aeration was the experimental reactor (micro-aeration), and the reactor without micro-aeration was set as the control. During the operation, mechanical stirring was applied at 100 rpm with an agitator. The ORP (oxidation–reduction potential) probe was used to monitor the variations of oxidation reduction potential in the reactor. Initially, 3 L of primary sludge was fed into the reactor, and then 1 L of primary sludge was discharged and fed daily. The liquid and gas samples were collected regularly for further analysis. The sludge samples on day 30 were collected for metagenomic and real-time quantitative PCR (RT-qPCR) analysis.

2.4. Analytical methods

Total solids (TS), volatile solids (VS), chemical oxygen demand (COD), NH_4^+-N, NO_3^--N, NO_2^--N, and PO_4^{3-}-P were measured according to standard methods [3]. The total organic carbon (TOC) was analyzed using a TOC analyzer (Shimadzu TOC-L CPH, Japan). Soluble polysaccharide content was determined by the anthrone-sulfuric acid method with glucose as the standard. Soluble protein concentration was determined using a BCA protein determination kit (Beijing Suolaibao Technology Co., Ltd). CH_4, NO_2, and NO contents in the gas phase were determined using a gas chromatography (GC1690, Hangzhou Jiedao, China) with a capillary column and a flame ionization detector (FID). VFAs were determined by a gas chromatography (GC7900, Tianmei, China) with a TM-FFAP column (30 m × 0.32 mm (ID) × 0.5 μm) column and a FID detector. The determination of enzyme activities and excitation emission matrix (EEM) fluorescence spectra was detailed in Text S1 (SI). Detailed extraction protocol of P was provided in Text S2 (SI).

2.5. Metagenomic sequencing, RT-qPCR and 16S rRNA gene sequencing

The metagenomic sequencing were conducted by Majorbio Biopharm Technology Co., Ltd (Shanghai, China). Samples were stored at −80 °C before test. The genomic DNA was extracted using an E.Z.N.A ®soil kit (Omega Bio tek, USA). The concentration and purity of obtained DNA samples was assessed using an ultra-micro spectrophotometer (NanoDrop2000, USA), and 1 % agarose gel electrophoresis. Extracted DNA was broken into fragments of around 400 bp, and sequencing library was constructed using the NEXTFLEX™ Rapid DNA-Seq Kit. Samples were sequenced on an Illumina platform with Illumina NovaSeq Reagent Kits. The obtained data was analyzed in the online Majorbio Cloud Platform (https://www.majorbio.com). RT-qPCR and 16S rRNA gene sequencing are detailed in Text S3 and Text S4 (SI).

3. Results and discussion

3.1. VFAs production and organics dissolution of the batch experiment

3.1.1. VFAs production from primary sludge

Variations of VFAs concentration and distribution during the anaerobic fermentation are shown in Fig. 1. In the control group (Fig. 1a), the VFAs concentrations increased initially and reached a peak of 2094.7 mg COD/L on day 6, and then gradually decreased to 1534.4 mg COD/L on day 9 possibly owing to the activities of methanogens [33]. In contrast, in the micro-aeration group, the VFA production reached the maximal value of 1662.5 mg COD/L on the first day, followed by a slow decrease (437.3 mg COD/L on day 9). This trend indicated that the micro-aeration posed adverse effects on VFAs accumulation during the anaerobic fermentation process. The consumption of VFAs further resulted in rising pH values, creating a suitable environment for methanogens to grow [23]. Therefore, VFAs would be further consumed over time to form CH_4 as time extended.

To further explore the fermentation performance, VFAs distribution and SCOD composition were explored. It was found that VFAs were the main organic component accounting for more than 57 % of the SCOD in the batch experiments (Fig. 1d), which was also observed in a previous study [25]. Variations of VFA distributions in the control and micro-aeration groups are illustrated in Fig. 1b, c. Acetate and propionate were the dominant VFAs produced during the fermentation in both groups, in agreement with previous literature [45]. In the control group, the acetate abundance increased to 53 % on day 4 and subsequently exhibited a gradual decline to 42 %, while propionate abundance fluctuated around 20 % during the whole reaction. In contrast, the acetate abundance decreased significantly from 48 % to around 27 % in the micro-aeration group. It might be attributed to that acetate was more easily utilized by bacteria [50]. As acetate decreased, the propionate abundance in this group gradually increased from 19 % to 32 % on day 4, and then began to decline to 15 % over time. The abundances of *iso*-valerate, butyrate, and *iso*-butyrate all showed an increase over time, which was possibly owing to the rapid consumption of acetate and propionate with micro-aeration.

3.1.2. Characteristics of the dissolved organic matters

Generally, the anaerobic fermentation includes hydrolysis, acidogenesis, acetogenesis, and methanogenesis, and hydrolysis is widely accepted as the rate-limiting step [12,56]. To understand how micro-aeration affects the dissolution of organics from sludge solids, the concentrations of carbohydrates and proteins in the soluble fraction were determined under different operation conditions (micro-aeration and control) (Fig. 2). The soluble carbohydrates accumulated during the first several days in both groups. In the control group (Fig. 2a), it increased from an initial 15.2 mg COD/L to 36.1 mg COD/L, followed by a rapid decrease from day 8. The introduction of micro-aeration significantly increased carbohydrate release, which achieved the maximal value (76.3 mg COD/L) on day 5 (Fig. 2b). Conversely, a different variation pattern was observed for protein concentrations over time. While the protein concentration reached its maximum of 1152.5 mg COD/L on day 6 in the control group, protein concentrations in the micro-aeration group peaked sooner (day 2) at lower values (692.0 mg COD/L). This might be attributed to released proteins being rapidly consumed by microorganisms. It can be concluded that the micro-aeration facilitated the release and conversion of the organic matters especially proteins.

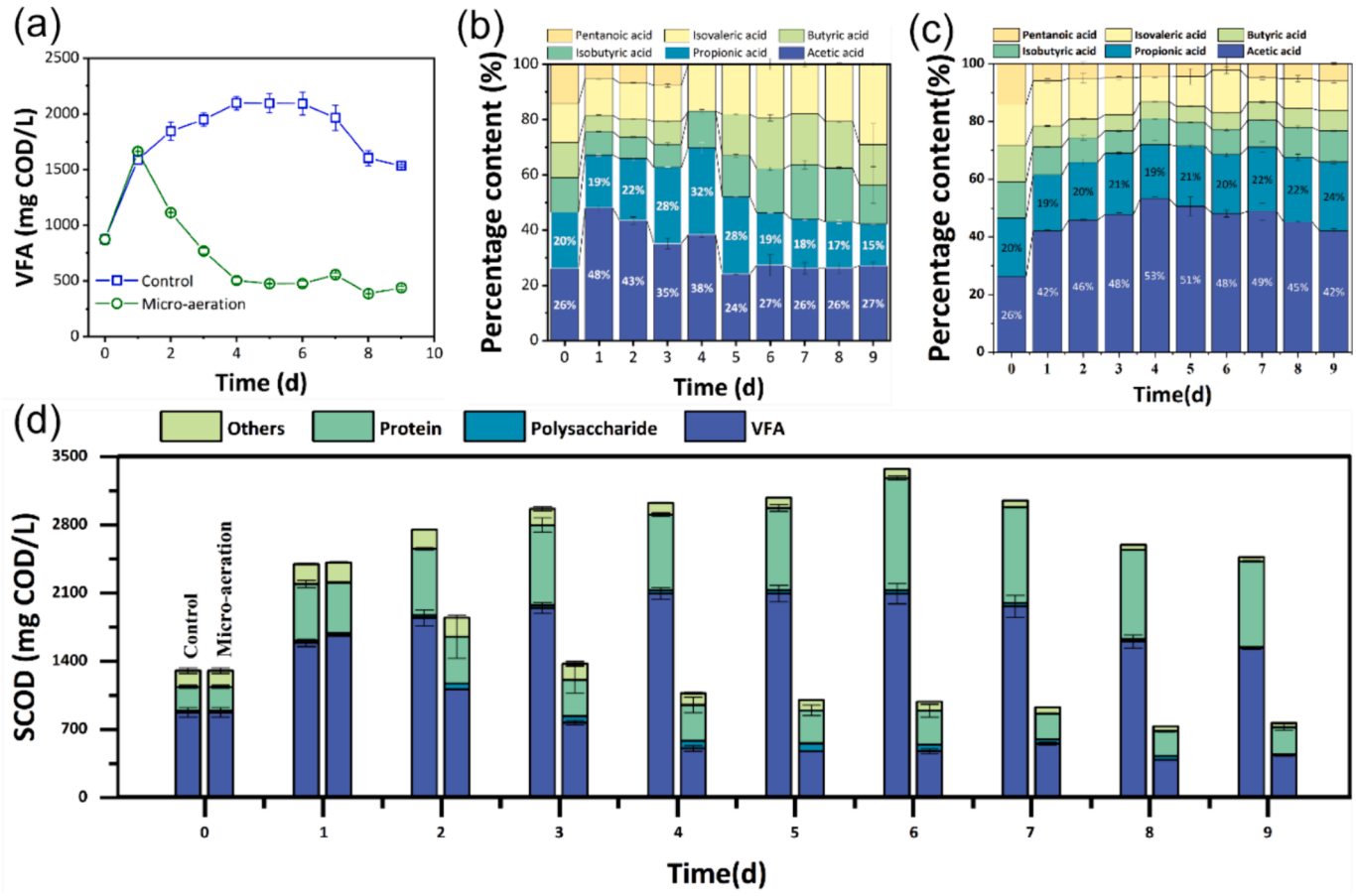


Fig. 1. VFA production and compositions during the fermentation process: (a) Total VFA yield; (b, c) VFA distribution in the micro-aeration and control groups; (d) SCOD compositions.

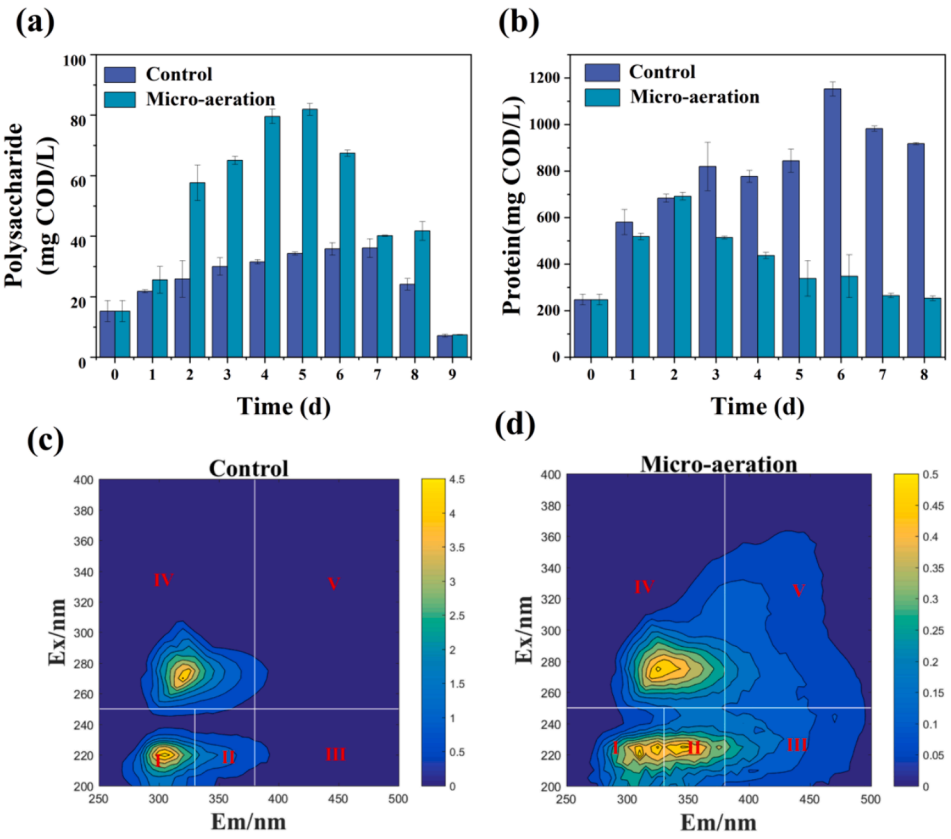


Fig. 2. The variations of soluble carbohydrate (a) and protein (b) in the supernatant. The supernatant EEM spectra of the control (c) and micro-aeration (d) groups.

To explore the spectral difference of solubilized organics between these two groups, EEM spectra of the supernatant on day 3 were obtained (Fig. 2c, d). The EEM spectrum was divided into five regions based on excitation wavelength (EX) and emission wavelength (EM) [5,13,44]. It was found that the main soluble organics with and without micro-aeration were mainly simple aromatic proteins (region I and II) and soluble microbial products (SMP, region IV). SMP are mainly composed of microbial metabolites such as carbohydrate and proteins, which were widely observed in anaerobic fermentation systems [52,55]. Fluorescence regional integration (FRI) based on EEM spectra (Fig. S2).

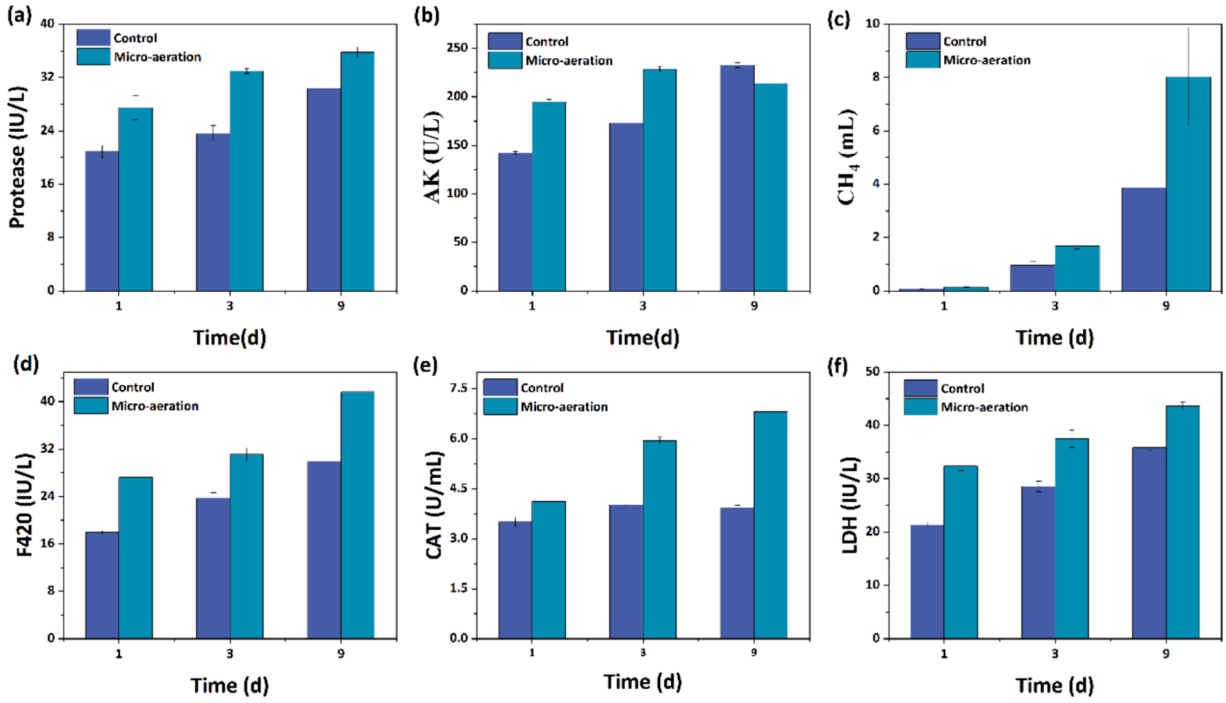


Fig. 3. Key enzymes activities and environment variables during the fermentation: (a) Protease (b) AK; (c) CH_4; (d) F420; (e) CAT and (f) LDH.

While region I and II accounted for most of the fluorescence signal in both control and the micro-aeration groups, the fluorescent abundance of region II was much higher in the micro-aeration group (48 %) than the control group (35 %). The fluorescence of region IV (SMP) also increased from 6 % (control) to 13 % (micro-aeration). These observations indicated that the micro-aeration induced enhanced solubilization of organics such as aromatic proteins (region II) and SMP (region IV) from the primary sludge [55].

3.2. Individual fermentation processes and N distribution of the batch experiment

3.2.1. Hydrolysis, acetogenesis and methanogenesis processes

Micro-aeration was reported to facilitate hydrolysis process in anaerobic fermentation [24]. It can also benefit methane production, resulting in strong and continuous VFA consumption [35]. Therefore, it is necessary to explore the role of micro-aeration in the hydrolysis, acetogenesis and methanogenesis steps. Considering that the protein was much higher than the carbohydrate (Fig. 1d), protease activity was chosen as a measure to reveal the activity of hydrolysis process [57]. In addition, activities of enzymes acetate kinase (AK) and coenzyme F420 (F420) were obtained to characterize the activities of the acetogenesis and methanogenesis processes, respectively (Fig. 3). The protease activity with micro-aeration was consistently higher than in the control group (Fig. 3a), indicating that the presence of oxygen did promote the hydrolysis process of primary sludge. This is consistent with the rapid decrease of soluble protein observed earlier (Fig. 2b). Similarly, the AK activities in the micro-aeration group were higher than the control group (on Day 1 and 3) (Fig. 3b), suggesting that the acetogenesis process was enhanced in the presence of tiny amounts of O_2. However, on day 9, the AK activity of the micro-aeration group was lower than in the control group, possibly due to insufficient substrates (i.e., proteins and carbohydrates). In general, these observations implied that micro-aeration was beneficial for promoting both the hydrolysis and acetogenesis process.

Both the methane production and F420 activity were monitored to reveal the impact of micro-aeration on the methane production. As depicted in Fig. 3c, the cumulative methane production in the micro-aeration group (8.03 mL) was much higher than the control group (3.87 mL). This, coupled with the reduced VFAs concentration (Fig. 1), revealing an enhancement of methanogenic activity due to micro-aeration. Additionally, the VFAs consumption would result in pH decrease over time, creating a more suitable environment for the methanogens [43]. Meanwhile, the F420 activities gradually increased over time in both groups, and the micro-aeration group showed a relatively higher F420 activity than that in the control group (Fig. 3d). Notably, F420 was a key enzyme involved in hydrogenotrophic methanogenesis [28], indicating that the hydrogenotrophic methanogenesis was also improved with micro-aeration. Collectively, the above observations implied that application of micro-aeration promoted all processes including hydrolysis, acetogenesis and methanogenesis.

3.2.2. Effect of micro-aeration on microbial activity

To further explore the influence of micro-aeration on microbial activities, catalase (CAT) and lactate dehydrogenase (LDH) activities during the fermentation were further determined. As demonstrated by the observation of CAT activities in Fig. 3e, the intervention of micro-aeration may lead to the production of reactive oxygen species (ROS), thus resulting in higher CAT activity in the presence of micro-aeration [54]. Similarly, the introduction of micro-aeration also increased the LDH activity in the reactor (Fig. 3f). As proper ROS level could stimulate the physiological activity [28,60], the ROS induced by micro-aeration might improve the microbial activity, thus contributing to enhanced anaerobic fermentation processes as discussed above.

3.2.3. Nitrogen distribution in the anaerobic system

The distribution of N during the fermentation was further investigated (Fig. S3a). The soluble NH_4^+-N in the control group gradually increased and reached a maximum value of 379.2 mg/L on day 9. However, little NO_3^--N or NO_2^--N was detected during the whole fermentation. Interestingly, the maximal concentration of soluble NH_4^+-N in the micro-aeration group (286.8 mg/L, day 9) was only about 75.6 % of that in the control group. A possible explanation may be that the introduction of O_2 could create aerobic-anoxic-anaerobic micro-environments within the serum bottles, resulting in simultaneous nitrification and denitrification [61]. After fermentation, the N distribution in the solid and liquid phase was determined (Fig. S3b). The solid phase N fraction after fermentation in both groups were significantly lower than the influent. This could be attributed to decomposition of organic substances (e.g., proteins) during which soluble NH_4^+-N was produced [41]. Since micro-aeration promoted the hydrolysis process, more N should be released from the solid phase of the sludge. Indeed, a lower solid phase N concentration was observed in micro-aeration group (611.1 mg/L) than that in the control group (672.8 mg/L). In addition, the micro-aeration group accumulated more NO_2^--N (3.81 mg/L) in the supernatant than the control group (0.14 mg/L). It was reported that NO_2^--N usually accumulated attributing to electron competition between nitrate reductase and nitrite reductase [11]. Hence, the higher NO_2^--N and NO_3^--N level further support the hypothesis that micro-aeration promoted simultaneous nitrification and denitrification in this system.

Theoretically, 2.86 mg COD is required to remove 1 mg NO_3^--N during denitrification [48]. At the end of fermentation, the total N concentration (i.e., NO_3^--N + NO_2^--N + NH_4^+-N + TN-solid) in micro-aeration group was 131.6 mg/L lower than in the control group (Fig. S3b). Assuming that all 131.6 mg/L of N was converted to NO_3^--N, the denitrification process of those NO_3^--N would theoretically require 376.4 mg COD/L carbon source. However, compared with the control group, the VFA and SCOD in micro-aeration group were reduced by 832.1 mg COD/L and 1701.8 mg/L, respectively (Fig. 1). Hence, a large faction of VFA and SCOD may be consumed by other microbial activities such as methane production and aerobic respiration.

3.3. Anaerobic fermentation performance in semi-continuous reactors

3.3.1. VFAs production performance

To evaluate the effect of micro-aeration on VFA accumulation under semi-continuous mode, VFAs concentrations were measured over time (Fig. 5). In the batch experiment above, it was observed that CH_4 production in the micro-aeration group was significantly increased over the control group (Fig. 3f). Shorter SRTs are commonly regarded as more conducive for VFA accumulation by mitigating the influence of methanogens [1,46]. Therefore, a short SRT (3 d) was optimized in advance and adopted in the semi-continuous experiment. VFA production in the semi-continuous reactors are illustrated in Fig. 4. The VFA production in both reactors increased at first, and decreased after day 10 onwards due to fluctuations of the influent (Fig. 4a), and finally re-increased gradually until stabilizing from around day 24. During the steady stage, the VFA concentration in the micro-aeration group reached 1343.6 mg COD/L, while the control group reached 1645.8 mg COD/L. Compared to the influent, the effluent VFA in the micro-aeration and control groups increased by 91.0 % and 134.0 %, respectively (Fig. S4). Compared to the VFA reduction in batch experiment (832.1 mg COD/L), VFA in micro-aeration group was only about 300 mg COD/L lower than that in the control, indicating that short SRT might reduce VFA consumption by methanogens.

Moreover, soluble nitrogen and phosphorus during the operation were also monitored (Fig. 4b, c). Initially, the soluble PO_4^{3-}-P concentrations in both groups accumulated rapidly, and reached their peaks around 70 mg/L on day 3. This could be attributed to phosphate release during the fermentation. Then, the soluble PO_4^{3-}-P further declined to 39.8 mg/L and 33.4 mg/L on day 15 in the control and micro-aeration

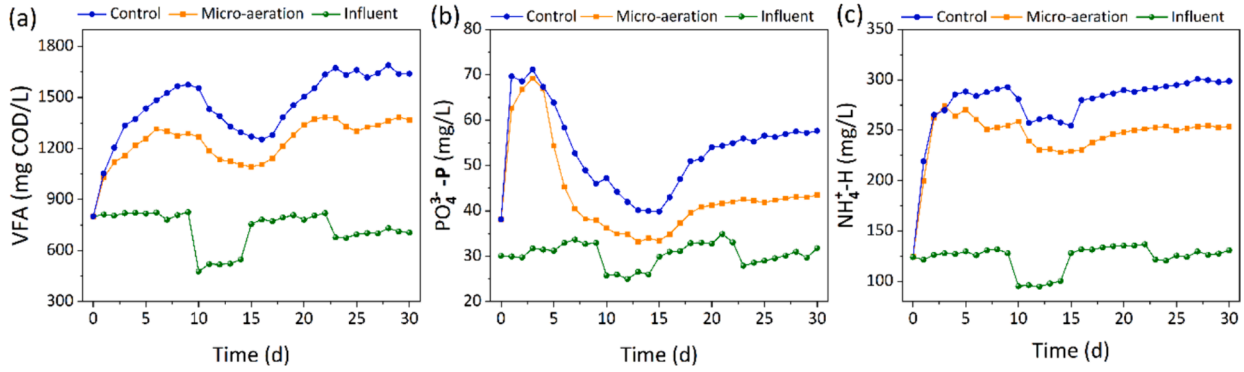


Fig. 4. Variations of (a) VFA, (b) phosphate, and (c) ammonia nitrogen during the semi-continuous fermentation process.

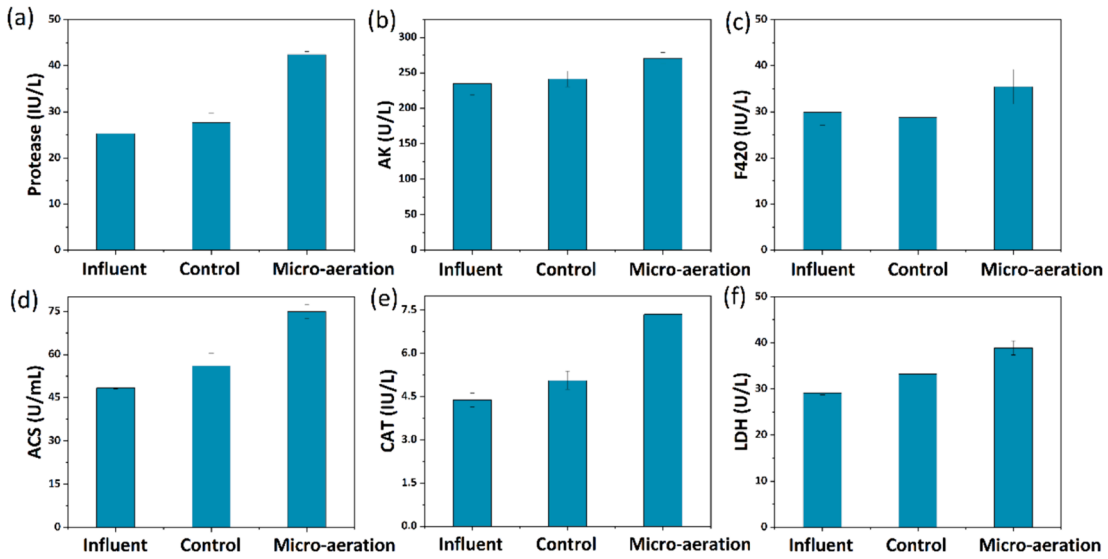


Fig. 5. Variations of enzyme activity during semi-continuous fermentation: (a) Proteinase; (b) acetate kinase; (c) F420; (d) ACS; (e) CAT; (f) LDH.

groups, respectively. Similarly, it also resulted from the decrease of influent PO_4^{3-}-P (Day 10–15). After that, the PO_4^{3-}-P in the control and micro-aeration groups increased gradually and stabilized at 57.7 mg/L and 43.5 mg/L, respectively. The soluble NH_4^+-N concentrations followed a similar trend to the PO_4^{3-}-P concentration in the steady state (Fig. 4c). The control group displayed a higher NH_4^+-N concentration (around 300 mg/L) than the micro-aeration group (around 250 mg/L), which was consistent with the batch experimental results. These results implied that micro-aeration could reduce both soluble NH_4^+-N and PO_4^{3-}-P in the fermentation broth.

3.3.2. *Key enzymes and microbial activities*

The activities of key enzymes were determined on day 30 when the semi-continuous reactors were operated stably (Fig. 5). The activities of protease and AK in the micro-aeration group increased by 53.3 % and 12.1 % respectively compared to the control group (Fig. 5a, b), indicating enhanced hydrolysis and acetogenesis activities. As for methanogenesis, F420 and acetyl-coenzyme A synthase (ACS) are key enzymes in hydrogenotrophic and acetoclastic methanogenesis, respectively [28]. During the acetoclastic methanogenesis, acetate is first converted into acetyl-coenzyme A, followed by ACS dependent hydroxylation, a key step to transform acetate into methane [8]. As shown in Fig. 5c, d, micro-aeration improved both the activities of F420 and ACS, with the activities increased by 23.3 % (F420) and 33.8 % (ACS) compared to the control. It clearly implied that micro-aeration can promote both the hydrogenotrophic and acetoclastic methanogenesis, which was also observed in the batch experiment. Moreover, the activities of CAT and LDH (Fig. 5e, f) were also improved similar to the results obtained in the batch experiments. Together with the high functional enzymes' activities, it indicated that the micro-aeration stimulated and enhanced the overall microbial activities.

3.3.3. *Carbon, nitrogen and phosphorus balance*

After the reactor was stabilized, carbon and nitrogen balances were established based on their total content in the corresponding reactor (Fig. 6). The carbon content in the solid, liquid, and gas phases of sludge was shown in Fig. 6a. It was found that the majority of carbon was still present in the solid phase. In the micro-aeration group, solids accounted only 63 % of the carbon content (80 % in the control), revealing that micro-aeration did promote the solubilization of the solids. The carbon contents in the liquid phase and CH_4 were similar in both groups, further implied that VFA consumption by methanogens was alleviated. Interestingly, the carbon content distributed in "others" with micro-aeration (18 %) was much higher than that in the control (2 %), which may be due to respiration by faculty bacteria and denitrification [10]. As for nitrogen (Fig. 6b), the introduction of micro-aeration resulted in more N released from the solid phase, and TN concentrations in the solid phase was 558.0 mg/L (control) and 521.7 mg/L (micro-aeration), respectively. However, the liquid phase NH_4^+-N concentration of 297.4 mg/L in the control group was higher than that in the micro-aeration group (251.7 mg/L). Similar to the batch experiment, the slight accumulation of NO_2^--N indicated that nitrification and denitrification were promoted

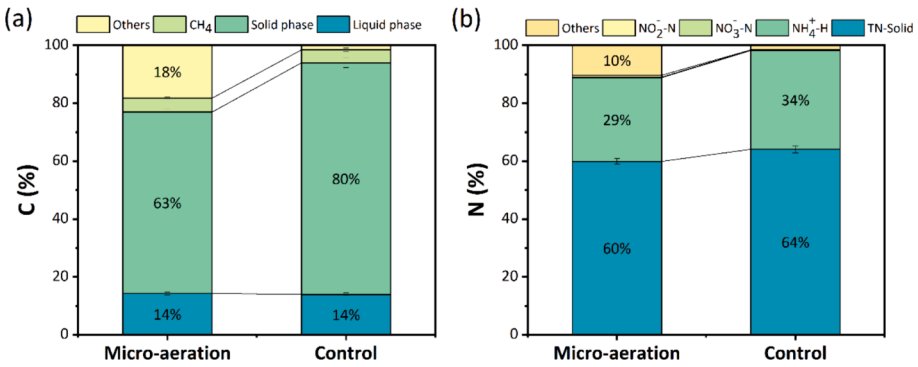


Fig. 6. Balance analysis for (a) carbon and (b) nitrogen. The total carbon and nitrogen content in the influent were set as 100%.

by micro-aeration. This can also partly explain the higher carbon (18 %) and nitrogen (10 %) distribution in "others" for the micro-aeration group.

As for phosphorus, the total phosphorus (TP) amount was similar in the control (476.5 mg) and micro-aeration (470.9 mg) reactors (Fig. S6). However, the TP in the solid phase was much higher in the micro-aeration reactor (340.4 mg) than in the control reactor (303.4 mg), in agreement with the lower soluble PO_4^{3-}-P in the micro-aeration group (Fig. 4b). The sequential extraction was conducted to further reveal the variations of P distribution in the solid phase. It was found that increased organic-P in solids (42 % in micro-aeration versus 39 % in control) mainly accounted for the P increase in sludge solids. The above results indicated that application of micro-aeration didn't change the TP but the distribution of P.

3.4. Microbial characteristics of the semi-continuous reactor

3.4.1. Microbial community structure

The microbial community structure at phylum and genus levels are further depicted in Fig. 7. As shown in Fig. 7**a**, Phyla *Actinobacteriota*, *Proteobacteria*, *Firmicutes*, and *Chloroflexi* were the main phyla which were commonly detected in anaerobic reactors [15]. Among them, the VFA consuming Phylum *Proteobacteria* increased from 31.53 % (control) to 43.59 % (micro-aeration), which explains the relatively low VFA content in the micro-aeration group [16]. Phylum *Firmicutes* relating to hydrolysis and acidification increased from 10.02 % (control) to 12.96 % (micro-aeration) [62]. The bacterial community at genus level was shown in Fig. 7b. Genus *Acinetobacter* relating to hydrolysis was the most dominant genus in micro-aeration reactor with a relative abundance of 35.44 % (versus 19.92 % in the control) [40]. In addition, some species from this genus was also reported to be polyphosphate accumulating organisms (PAOs) [22]. Another dominant genus *Candidatus_Microthrix* relating to removal of organic pollutants, TN, and phosphate accounted for abundances of 16.45 % (control) and 14.4 % (micro-aeration), respectively [27,32]. The distribution of archaea is depicted in Fig. 7c. Acetoclastic methanogens are mainly composed of *Methanosarcina* and *Methanosaeta* [28,42], *Methanosaeta* occupied similar abundances in the control (30.72 %) and micro-aeration (28.79 %) groups. However, higher relative abundance of hydrogenotrophic *Methanobrevibacter* was observed in micro-aeration group (32.31 %) than the control (17.75 %), indicating that micro-aeration enriched the methanogens capable of utilizing H_2.

3.4.2. RT-qPCR of key functional genes

The key functional enzyme genes abundance involved in VFAs production and methane production was also determined (Fig. S7). Among

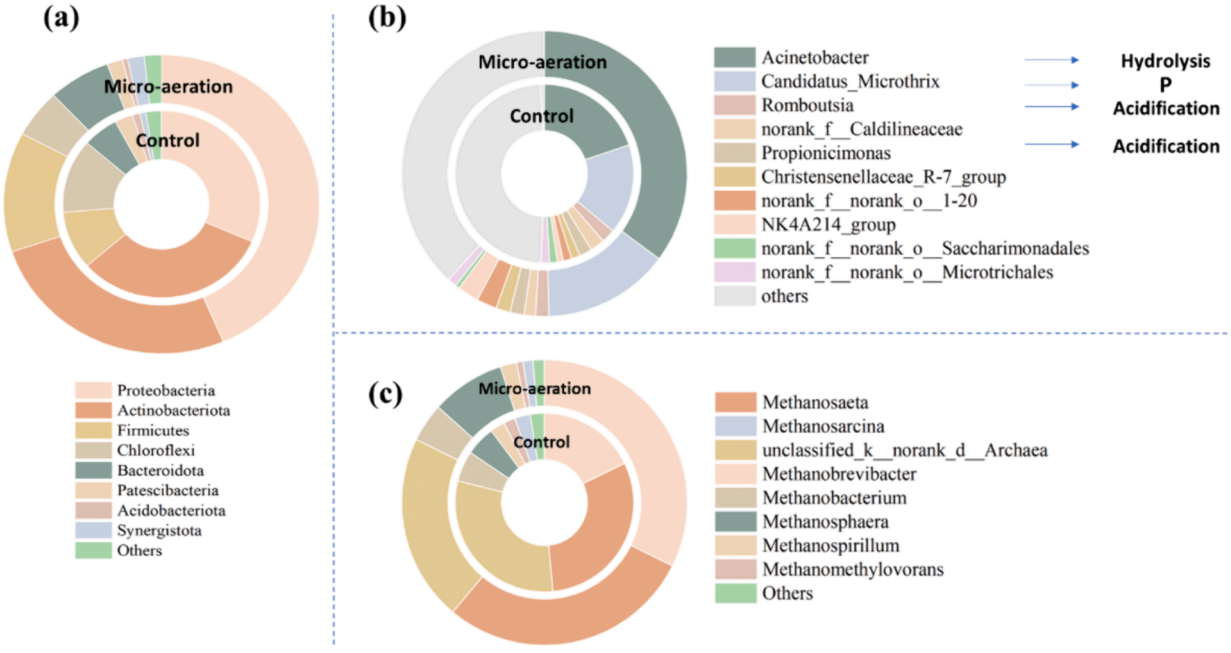


Fig. 7. Microbial community structure of the fermentation reactor: (a) Bacterial distribution at phylum level; (b) bacterial distribution at genus level; (c) distribution of archaea at genus level.

them, FTHFS, mcrA and ACS genes are associated with homoacetogenesis, methanogenesis and acetoclastic methanogenesis, respectively [18]. In the presence micro-aeration, the abundance of FTHFS genes increased to 8.09×10^5 copies/g from 3.52×10^5 copies/g in the control. This result again supported that micro-aeration can promote the acetogenesis activity. However, the high abundance of mcrA gene further implied that the introduction of micro-aeration is also conducive to methane production. Specifically, the abundance of ACS gene in the micro-aeration group (5.68×10^7 copies/g) was much higher than that in the control group (2.34×10^7 copies/g). It indicated that both acetoclastic and hydrogentrophic methanogenesis were improved by micro-aeration. The higher ACS enzyme activity (Fig. 5d) also supported this conclusion. To explain the nitrogen transformation in the anaerobic fermentation system, functional genes relating to denitrification were then analyzed. The narG gene (Fig. S7d) plays a crucial role in reducing nitrate to nitrite during denitrification. It could be seen that the micro-aeration almost increased the abundance of this gene by 100 % (9.20×10^7 copies/g) compared to the control group (4.54×10^7 copies/g). It verified the occurrence of denitrification induced by micro-aeration and explained the TN reduction in the nitrogen balance discussed earlier.

3.4.3. Metagenomic analysis

During anaerobic fermentation, micro-aeration might induce ROS production, which might improve the microbial activity as previously discussed [60]. The metagenomic results also supported this speculation. During the hydrolysis process, the gene abundances of protease (EC: 3.4.21.53) and α-glucosidase (EC: 3.2.1.20) in micro-aeration Group were 123.79 % and 105.90 % of those in the control group, respectively. This was in accordance with the higher hydrolysis enzymes activities (Fig. 5a). After hydrolysis, the produced low molecular weight organics would be further bio-converted into acetate by AK [23]. The abundance of AK (EC: 2.7.2.1) gene in the micro-aeration group was about 121.40 % of that in the control group. Clearly, the acetogenesis activity was also improved by micro-aeration, consistent with the AK activity (Fig. 5b).

As for methane production, the functional genes in all the methanogenesis pathways exhibited in KEGG database were detected (Fig. 8). The methyl-CoM reductase gene involved in the last step of CH_4

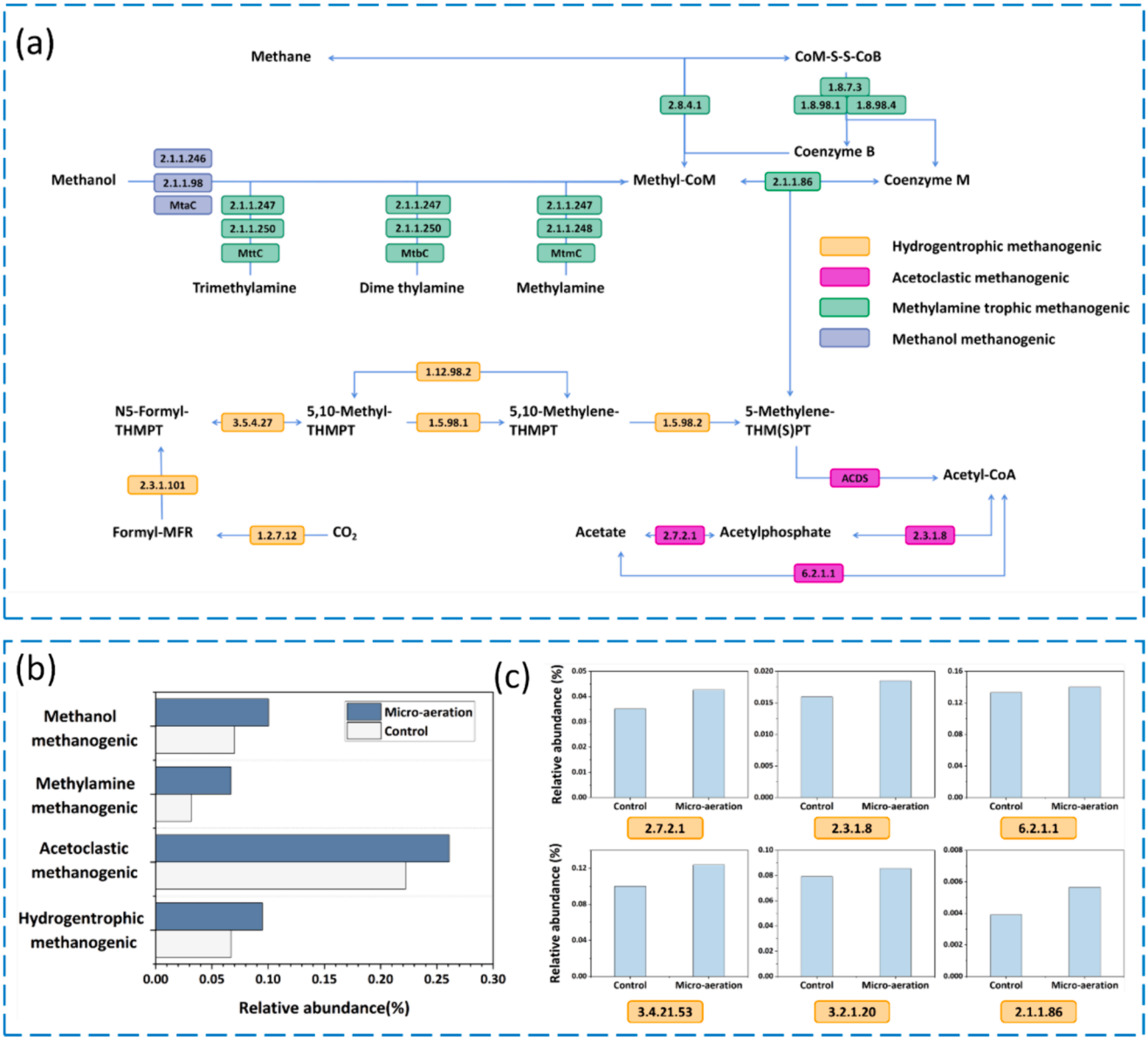


Fig. 8. (a) Methanogenesis pathway; (b) the relative gene abundances of different methanogenesis pathways; (c) the relative abundances of functional enzyme genes in hydrolysis, acetate production and methanogenesis.

formation was significantly upregulated in the micro-aeration group (187 % higher than control) [28]. All methanogenesis pathways were enhanced by micro-aeration (Fig. 8b). The relative abundance of hydrogenotrophic pathway increased by 41.2 % compared to the control, which was in consistent with the enriched *Methanobrevibacter*. However, the acetoclastic methanogenesis related genes still took the most dominant place. During this pathway, the acetyl-CoA would be formed with acetyl-CoA synthetase (EC: 6.2.1.1) or phosphate acetyltransferase (EC: 2.3.1.8). Methyl-CoM would be further formed by tetrahydromethanopterin methyltransferase (EC: 2.1.1.86) [8]. In the

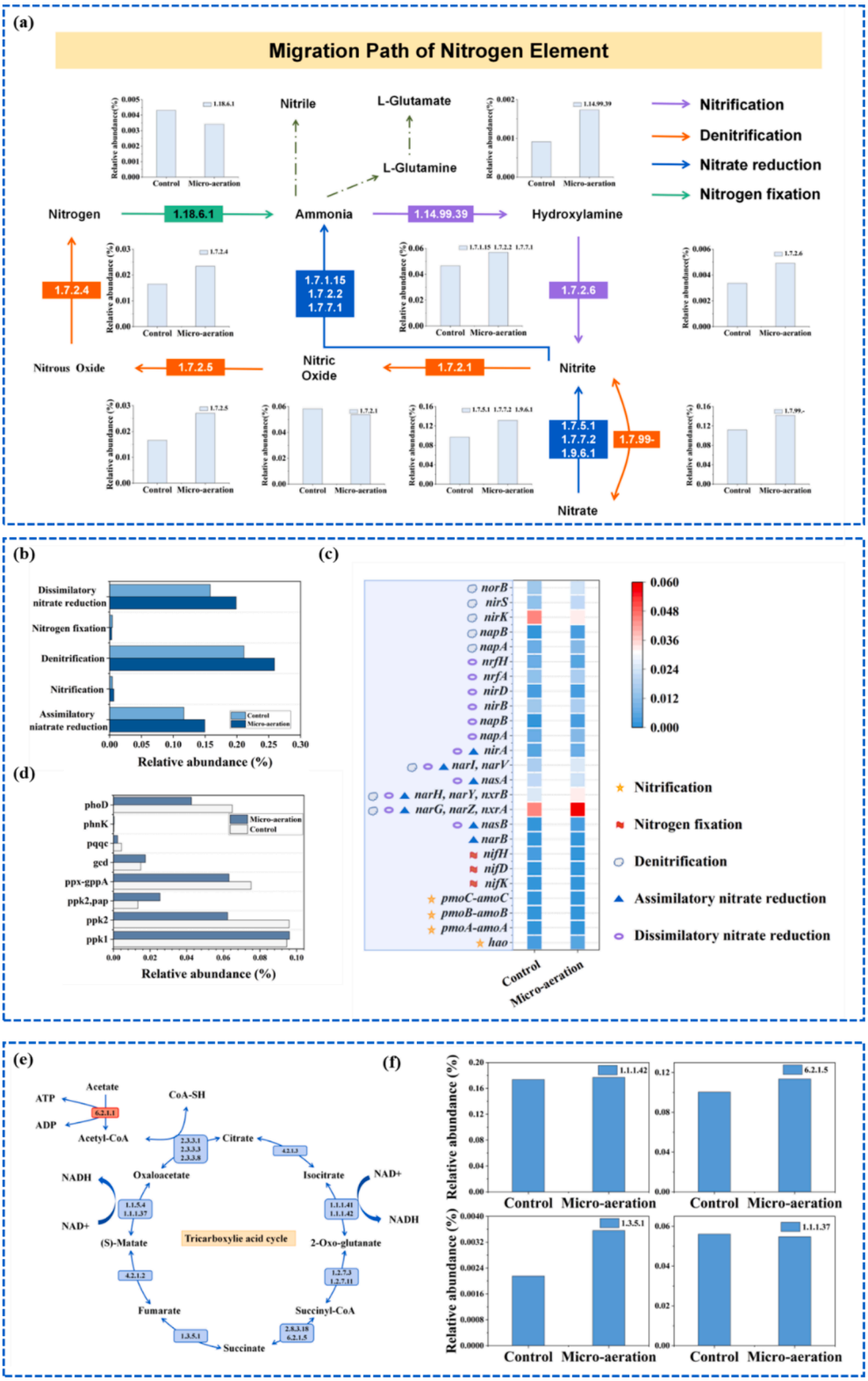


Fig. 9. (a) Nitrogen metabolic pathway and the relative abundances of key functional genes, (b) the relative abundances of key genes affiliated to different nitrogen metabolic pathways, (c) relative abundance heatmap of key nitrogen metabolic genes, (d) the relative abundance of key genes relating to phosphorus, (e) schematic diagram of the tricarboxylic acid (TCA) cycle pathway and (f) the relative abundances of key genes of TCA cycle.

micro-aeration group, the abundances of these enzymes' genes were 105.17 % (EC: 6.2.1.1), 115.92 % (EC: 2.3.1.8), and 144.87 % (EC: 2.1.1.86) of those in the control group, respectively. These results implied that micro-aeration did enhance the CH_4 production with acetoclastic methanogenesis being the dominant pathway, which would contribute to consumption of the accumulated VFAs.

3.5. N and P transformation pathway in the semi-continuous reactor

During the fermentation, N and P from solid phase would be released into the liquid phase (Fig. 4), which might pose negative effects on the subsequent process. Hence, the N and P transformation pathways in current systems were then analyzed. Based on the metagenomic results, both nitrification and denitrification genes were detected in the reactors. It indicates the possibility of simultaneous occurrence of nitrification and denitrification processes. On the one hand, the abundances of nitrification genes increased in the presence of micro-aeration (Fig. 9a). Compared to the control group, the abundances of ammonia monooxygenase (EC: 1.14.99.39) and hydroxylamine dehydrogenase (EC: 1.7.2.6) genes increased by 89.72 % and 46.13 %, respectively. This result could explain the relatively lower ammonia (Fig. 4c) and higher NO_2^--N (Table S6) concentrations in the micro-aeration group than the control group. However, similar gene abundances of nitrite oxidoreductase (EC: 1.7.99.-) were observed in both reactors. On the other hand, the denitrification genes were also detected with higher abundances in the micro-aeration group. For example, the abundances of functional genes encoding nitric oxide reductase (EC: 1.7.2.5), and nitrous oxide reductase (EC: 1.7.2.4) in the micro-aeration group increased by 63.45 %, and 41.91 % compared to the control group, respectively. It indicated that micro-aeration also enhanced the denitrification process which decreased the total nitrogen content of the system.

To better exhibit the effect of micro-aeration on nitrogen metabolism, the genes' abundances affiliated to different nitrogen metabolism pathway are illustrated in Fig. 9b, c. The nitrification and denitrification genes displayed higher abundances among these pathways, and both pathways' genes were upregulated by micro-aeration. From another point of view, the tricarboxylic acid (TCA) cycle is the central metabolic pathway of carbon, through which the electrons are mainly produced [51]. It is also regarded as the energy center of denitrifying bacteria, providing the reducing power of the electron transfer chain and the synthetic metabolic substrates for cell growth [14,26]. Thus, the genes encoding TCA enzymes were further investigated (Fig. 9e, f). Within this pathway, the abundance of gene encoding succinate dehydrogenase (EC: 1.3.5.1) increased by 65.37 % with micro-aeration compared to the control group, indicating higher activities of carbon metabolism and reducing power production. Considering the relatively lower VFA concentration and upregulated denitrification pathway with micro-aeration, it further indicated that the micro-aeration might improve VFA consumption via TCA cycle, providing more reducing power for microbial denitrification.

As for the phosphorus metabolism, polyphosphate kinase 1 (PPK1) and polyphosphate kinase 2 (PPK2) genes were the dominant functional genes [7,20]. Specifically, PPK1 is involved in polyphosphate (poly-P) formation in PAOs, while PPK2 hydrolyzes poly-P to synthesize GTP (Guanosine triphosphate) [7,36]. The relative abundance of PPK1 gene wasn't impacted much by the micro-aeration, while that of PPK2 gene decreased by 35.0 % compared to the control group. The phosphate hydrolase (PPX) is another enzyme responsible for the hydrolysis of poly-P (Fig. 9d) [31,36]. The PPX gene encoding this enzyme was also down-regulated by the micro-aeration (0.063 % vs. 0.075 % in the control). It indicated that the degradation of poly-P was inhibited by micro-aeration, which was in accordance with the observed high P content in the solid phase. Moreover, the alkaline phosphatase D (phoD) gene is related to organic-P mineralization, and the abundance of this gene was significantly reduced by micro-aeration (0.043 % vs. 0.065 % in control). Considering the higher organic-P distribution in solid phase, these results indicated that micro-aeration inhibit both extracellular organic-P mineralization and intracellular poly-P degradation, resulting in the lower soluble PO_4^{3-}-P.

Overall, the soluble NH_4^+-N and PO_4^{3-}-P concentrations in the fermentation broth were significantly reduced by micro-aeration. Considering that micro-aeration is regarded as a simple and economical approach that could be used in anaerobic system [10], engineering application of this technology is also expected to be technically feasible. However, most studies relating to micro-aeration is still limited to the laboratory scale [23,52], and the economic evaluation of micro-aeration technology is still lacking [10]. Hence, the anaerobic fermentation performance as well as economic characteristics adopting this technology in pilot and full scale still have to be comprehensively evaluated in the future study. Besides, several other limitations and challenges also need to be addressed before future industrial applications of this technology. For example, different influence pattern of micro-aeration on anaerobic fermentation were observed in different studies [35,52]. Hence, the oxygen dosage and micro-aeration pattern need to be further optimized for higher production of VFA and better control of soluble N and P. Considering the low solubility and poor mass transfer of oxygen into liquid phase, development of an efficient oxygen dosing method is also anticipated. In addition, hydrodynamics and mass transport will vary greatly during the scaling up process compared to the laboratory scale reactors. Hence, some other methods including hydrodynamics modeling should be incorporated for reactor design and optimization during further industrial applications.

4. Conclusion

This study systematically investigated the influence of micro-aeration on anaerobic VFAs production from primary sludge as well as the simultaneous N and P transformations. The specific conclusions are follows:

(1) The micro-aeration exhibited a non-targeted improvement effect on the anaerobic fermentation of primary sludge, including hydrolysis, acetogenesis, and methanogenesis processes. The overall microbial activity was also enhanced by micro-aeration as evidenced by the higher functional enzymes' activities.
(2) The micro-aeration enriched hydrogenotrophic genus *Methanobrevibacter* of the archaeal community and genus *Acinetobacter* of the bacterial community. However, the acetoclastic methanogenesis was still the dominant CH_4 production pathway that consumed VFA at the gene level.
(3) Compared to control, the application of micro-aeration could remove 131.6 mg/L more N in the system, and soluble NH_4^+-N was also reduced by it during the anaerobic fermentation. The occurrence of simultaneous nitrification and denitrification was found to be responsible for the total N loss of this system.
(4) The total P content in the reactor remained unchanged, while more P was distributed in the organic-P within the sludge solids (1.91 mg/g vs. 1.56 mg/g in the control). It could be attributed to that the micro-aeration inhibited both extracellular organic-P mineralization and intracellular poly-P degradation, resulting in more P retained in the solids phase.

Overall, this study demonstrated that micro-aeration could induce simultaneous transformation of soluble N and P during the anaerobic fermentation of primary sludge. These results could help to inform future design of fermentation processes, and to optimize for both N and P release using micro-aeration.

CRediT authorship contribution statement

Caihong Liu: Writing – review & editing, Writing – original draft.

Tanglong Zhang: Writing – original draft, Investigation. **Lin Li:** Writing – review & editing, Writing – original draft, Conceptualization. **Shaowu Liu:** Resources. **Jingjun Bo:** Visualization, Formal analysis. **Mohammed Mouhib:** Writing – review & editing. **Qiang He:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

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References

[1] D.E. Algapani, W. Qiao, M. Su, F. di Pumpo, S.M. Wandera, F. Adani, R.J. Dong, Bio-hydrolysis and bio-hydrogen production from food waste by thermophilic and hyperthermophilic anaerobic process, Bioresour. Technol. 216 (2016) 768–777.
[2] Y. An, M.Y. Wang, Z. Zhou, X. Sun, C. Cheng, S.M. Zhu, L.C. Wang, Z.C. Wu, Enhancing biodegradability of industrial park wastewater by packing carriers and limited aeration in the hydrolysis process, J. Clean Prod. 264 (2020) 10.
[3] APHA, A.P.H.A. 2005 Standard Methods for Examination of Water and Wastewater, 21st ed., Washington, DC, 2005.
[4] H. Chen, X.H. Zhao, Y.Y. Cheng, M.J. Jiang, X. Li, G. Xue, Iron robustly stimulates simultaneous nitrification and denitrification under aerobic conditions, Environ. Sci. Technol. 52 (3) (2018) 1404–1412.
[5] W. Chen, P. Westerhoff, J.A. Leenheer, K. Booksh, Fluorescence excitation – Emission matrix regional integration to quantify spectra for dissolved organic matter, Environ. Sci. Technol. 37 (24) (2003) 5701–5710.
[6] Y.D. Chen, H.F. Yang, H.J. Zou, T. Sun, M.X. Li, J. Zhai, Q. He, L. Gu, W.Z. Tang, Effects of acid/alkali pretreatments on lignocellulosic biomass mono-digestion and its co-digestion with waste activated sludge, J. Clean Prod. 277 (2020) 13.
[7] Y.M. Chuang, D.A. Belchis, P.C. Karakousis, The polyphosphate kinase gene *ppk2* is required for *Mycobacterium tuberculosis* inorganic polyphosphate regulation and virulence, mBio 4 (3) (2013).
[8] X. Duan, J.Y. Luo, Y. Su, C. Liu, L.Y. Feng, Y.G. Chen, Proteomic profiling of robust acetoclastic methanogen in chrysene-altered anaerobic digestion: Global dissection of enzymes, Water Res. 233 (2023) 10.
[9] W. Fang, X.D. Zhang, P.Y. Zhang, J.J. Wan, H.X. Guo, D.S.M. Ghasimi, X.C. Morera, T. Zhang, Overview of key operation factors and strategies for improving fermentative volatile fatty acid production and product regulation from sewage sludge, J. Environ. Sci. 87 (2020) 93–111.
[10] S.F. Fu, S.J. Lian, I. Angelidaki, R.B. Guo, Micro-aeration: an attractive strategy to facilitate anaerobic digestion, Trends Biotechnol. 41 (5) (2023) 714–726.
[11] L.X. Gong, M.X. Huo, Q. Yang, J. Li, B. Ma, R.L. Zhu, S.Y. Wang, Y.Z. Peng, Performance of heterotrophic partial denitrification under feast-famine condition of electron donor: A case study using acetate as external carbon source, Bioresour. Technol. 133 (2013) 263–269.
[12] A. Gonzalez, A. Hendriks, J.B. van Lier, M. de Kreuk, Pre-treatments to enhance the biodegradability of waste activated sludge: Elucidating the rate limiting step, Biotechnol. Adv. 36 (5) (2018) 1434–1469.
[13] Y.D. Guo, L. Guo, M. Sun, Y.G. Zhao, M.C. Gao, Z.L. She, Effects of hydraulic retention time (HRT) on denitrification using waste activated sludge thermal hydrolysis liquid and acidogenic liquid as carbon sources, Bioresour. Technol. 224 (2017) 147–156.
[14] Y. He, H. Yun, L. Peng, J. Ji, W.X. Wang, X.K. Li, Deciphering the potential role of quorum quenching in efficient aerobic denitrification driven by a synthetic microbial community, Water Res. 251 (2024) 12.
[15] Z.W. He, W.Z. Liu, C.C. Tang, B. Ling, Z.C. Guo, L. Wang, Y.X. Ren, A.J. Wang, Performance and microbial community responses of anaerobic digestion of waste activated sludge to residual benzalkonium chlorides, Energy Conv. Manag. 202 (2019) 9.
[16] Z.W. He, C.C. Tang, W.Z. Liu, Y.X. Ren, Z.C. Guo, A.J. Zhou, L. Wang, C.X. Yang, A. J. Wang, Enhanced short-chain fatty acids production from waste activated sludge with alkaline followed by potassium ferrate treatment, Bioresour. Technol. 289 (2019) 9.
[17] R. Hreiz, M.A. Latifi, N. Roche, Optimal design and operation of activated sludge processes: State-of-the-art, Chem. Eng. J. 281 (2015) 900–920.
[18] Z.S. Hua, Y.L. Wang, P.N. Evans, Y.N. Qu, K.M. Goh, Y.Z. Rao, Y.L. Qi, Y.X. Li, M. J. Huang, J.Y. Jiao, Y.T. Chen, Y.P. Mao, W.S. Shu, W. Hozzein, B.P. Hedlund, G. W. Tyson, T. Zhang, W.J. Li, Insights into the ecological roles and evolution of methyl-coenzyme M reductase-containing hot spring Archaea, Nat. Commun. 10 (2019) 11.
[19] J.G. Jiang, C.X. Gong, J.M. Wang, S.C. Tian, Y.J. Zhang, Effects of ultrasound pre-treatment on the amount of dissolved organic matter extracted from food waste, Bioresour. Technol. 155 (2014) 266–271.
[20] D.S. Jones, B.E. Flood, J.V. Bailey, Metatranscriptomic insights into polyphosphate metabolism in marine sediments, Isme J. 10 (4) (2016) 1015–1019.
[21] M. Layer, M.G. Villodres, A. Hernandez, E. Reynaert, E. Morgenroth, N. Derlon, Limited simultaneous nitrification-denitrification (SND) in aerobic granular sludge systems treating municipal wastewater: Mechanisms and practical implications, Water Res. X 7 (2020) 13.
[22] H.K. Li, Y.M. Zhong, H. Huang, Z.X. Tan, Y. Sun, H. Liu, Simultaneous nitrogen and phosphorus removal by interactions between phosphate accumulating organisms (PAOs) and denitrifying phosphate accumulating organisms (DPAOs) in a sequencing batch reactor, Sci. Total Environ. 744 (2020) 9.
[23] W. Li, Y.L. Liu, B.C. Wu, L. Gu, R. Deng, Upgrade the high-load anaerobic digestion and relieve acid stress through the strategy of side-stream micro-aeration: biochemical performances, microbial response and intrinsic mechanisms, Water Res. 221 (2022) 13.
[24] J.W. Lim, J.A. Chiam, J.Y. Wang, Microbial community structure reveals how microaeration improves fermentation during anaerobic co-digestion of brown water and food waste, Bioresour. Technol. 171 (2014) 132–138.
[25] H.B. Liu, H. Xiao, B. Yin, Y.P. Zu, H. Liu, B. Fu, H.J. Ma, Enhanced volatile fatty acid production by a modified biological pretreatment in anaerobic fermentation of waste activated sludge, Chem. Eng. J. 284 (2016) 194–201.
[26] S.Q. Liu, C. Wang, J. Hou, P.F. Wang, L.Z. Miao, Effects of Ag NPs on denitrification in suspended sediments via inhibiting microbial electron behaviors, Water Res. 171 (2020) 11.
[27] W.L. Liu, C. Shen, C. Liu, S.J. Zhang, S.F. Hao, Y.Z. Peng, J. Li, Achieving stable mainstream nitrogen and phosphorus removal assisted by hydroxylamine addition in a continuous partial nitritation/anammox process from real sewage, Sci. Total Environ. 794 (2021) 10.
[28] X.R. Liu, Q. Lu, M.T. Du, Q. Xu, D.B. Wang, Hormesis-like effects of tetrabromobisphenol A on anaerobic digestion: responses of metabolic activity and microbial community, Environ. Sci. Technol. 56 (16) (2022) 11277–11287.
[29] X. Ma, J.J. Ye, L. Jiang, L. Sheng, J.Y. Liu, Y.Y. Li, Z.P. Xu, Alkaline fermentation of waste activated sludge with calcium hydroxide to improve short-chain fatty acids production and extraction efficiency via layered double hydroxides, Bioresour. Technol. 279 (2019) 117–123.
[30] N. Mahmoud, G. Zeeman, H. Gijzen, G. Lettinga, Anaerobic stabilisation and conversion of biopolymers in primary sludge – Effect of temperature and sludge retention time, Water Res. 38 (4) (2004) 983–991.
[31] H.G. Martín, N. Ivanova, V. Kunin, F. Warnecke, K.W. Barry, A.C. McHardy, C. Yeates, S.M. He, A.A. Salamov, E. Szeto, E. Dalin, N.H. Putnam, H.J. Shapiro, J. L. Pangilinan, I. Rigoutsos, N.C. Kyrpides, L.L. Blackall, K.D. McMahon, P. Hugenholtz, Metagenomic analysis of two enhanced biological phosphorus removal (EBPR) sludge communities, Nat. Biotechnol. 24 (10) (2006) 1263–1269.
[32] P. Maza-Marquez, A. Castellano-Hinojosa, A. Gonzalez-Martinez, B. Juarez-Jimenez, J. Gonzalez-Lopez, B. Rodelas, Abundance of total and metabolically active Candidatus Microthrix and fungal populations in three full-scale wastewater treatment plants, Chemosphere 232 (2019) 26–34.
[33] Y. Miron, G. Zeeman, J.B. Van Lier, G. Lettinga, The role of sludge retention time in the hydrolysis and acidification of lipids, carbohydrates and proteins during digestion of primary sludge in CSTR systems, Water Res. 34 (5) (2000) 1705–1713.
[34] P. Neumann, Z. Gonzalez, G. Vidal, Sequential ultrasound and low-temperature thermal pretreatment: Process optimization and influence on sewage sludge solubilization, enzyme activity and anaerobic digestion, Bioresour. Technol. 234 (2017) 178–187.
[35] D. Nguyen, Z.Y. Wu, S. Shrestha, P.H. Lee, L. Raskin, S.K. Khanal, Intermittent micro-aeration: New strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion, Water Res. 166 (2019) 10.
[36] M. Ni, Y. Chen, Y. Pan, Y. Huang, D.P. Li, L. Li, B. Huang, Z.W. Song, Study on community structure and metabolic mechanism of dominant polyphosphate-accumulating organisms (PAOs) and glycogen-accumulating organisms (GAOs) in suspended biofilm based on phosphate recovery, Sci. Total Environ. 815 (2022) 11.
[37] H. Pang, Y. Xu, Y. Zhang, Q. Wei, D. Xu, J. Liu, J. Lu, Endogenous biopolymer hydrolysis for enhancing short-chain fatty acids recovery from excess sludge: Combination of lysozyme-catalyzing and cation exchange resin-mediated metal regulation, Chemosphere 341 (2023) 140102.

[38] H.L. Pang, Q. Wei, Y.M. Xu, Y.Y. Zhang, D. Xu, J.X. Liu, J.G. He, J.S. Lu, Two wastes into one resource: Carbide slag-driven anaerobic fermentation of excess sludge towards short-chain fatty acids recovery, Chem. Eng. J. 479 (2024) 11.
[39] H.L. Pang, Y.M. Xu, R.Y. Ren, J.G. He, X.L. Pan, L. Wang, Enhanced anaerobic digestion of waste activated sludge by alkaline protease-catalyzing hydrolysis: Role and significance of initial pH adjustment, Chem. Eng. J. 467 (2023) 12.
[40] L.N. Pang, Y. He, X.N. Liu, J.J. Li, P. Yang, The role of a newly isolated strain Corynebacterium pollutisoli SPH6 in waste activated sludge alkaline fermentation, Chemosphere 241 (2020) 9.
[41] R. Rajagopal, D.I. Masse, G. Singh, A critical review on inhibition of anaerobic digestion process by excess ammonia, Bioresour. Technol. 143 (2013) 632–641.
[42] A.E. Rotaru, P.M. Shrestha, F.H. Liu, M. Shrestha, D. Shrestha, M. Embree, K. Zengler, C. Wardman, K.P. Nevin, D.R. Lovley, A new model for electron flow during anaerobic digestion: direct interspecies electron transfer to *Methanosaeta* for the reduction of carbon dioxide to methane, Energy Environ. Sci. 7 (1) (2014) 408–415.
[43] S.R. Shin, S. Im, A. Mostafa, M.K. Lee, Y.M. Yun, S.E. Oh, D.H. Kim, Effects of pig slurry acidification on methane emissions during storage and subsequent biogas production, Water Res. 152 (2019) 234–240.
[44] J. Tong, Y.G. Chen, Recovery of nitrogen and phosphorus from alkaline fermentation liquid of waste activated sludge and application of the fermentation liquid to promote biological municipal wastewater treatment, Water Res. 43 (12) (2009) 2969–2976.
[45] A.S. Ucisik, M. Henze, Biological hydrolysis and acidification of sludge under anaerobic conditions: The effect of sludge type and origin on the production and composition of volatile fatty acids, Water Res. 42 (14) (2008) 3729–3738.
[46] S.M. Wandera, M. Westerholm, W. Qiao, D.M. Yin, M.M. Jiang, R.J. Dong, The correlation of methanogenic communities' dynamics and process performance of anaerobic digestion of thermal hydrolyzed sludge at short hydraulic retention times, Bioresour. Technol. 272 (2019) 180–187.
[47] S.W. Wang, F. Ma, W.W. Ma, P. Wang, G. Zhao, X.F. Lu, Influence of temperature on biogas production efficiency and microbial community in a two-phase anaerobic digestion system, Water 11 (1) (2019) 13.
[48] X.X. Wang, J. Zhao, D.S. Yu, G.H. Chen, S.M. Du, J.Y. Zhen, M.F. Yuan, Stable nitrite accumulation and phosphorous removal from nitrate and municipal wastewaters in a combined process of endogenous partial denitrification and denitrifying phosphorus removal (EPDPR), Chem. Eng. J. 355 (2019) 560–571.
[49] Y. Wang, X. Wang, D. Wang, T. Zhu, Y. Zhang, H. Horn, Y. Liu, Ferrate pretreatment-anaerobic fermentation enhances medium-chain fatty acids production from waste activated sludge: Performance and mechanisms, Water Res. 229 (2023) 119457.
[50] Y.Y. Wang, Y.L. Zhang, J.B. Wang, L. Meng, Effects of volatile fatty acid concentrations on methane yield and methanogenic bacteria, Biomass Bioenerg. 33 (5) (2009) 848–853.
[51] Q. Wei, J.S. Zhang, F.Z. Luo, D.H. Shi, Y.C. Liu, S. Liu, Q. Zhang, W.Z. Sun, J. L. Yuan, H.T. Fan, H.C. Wang, L. Qi, G.H. Liu, Molecular mechanisms through which different carbon sources affect denitrification by *Thauera linaloolentis*: Electron generation, transfer, and competition, Environ. Int. 170 (2022) 13.
[52] Y.X. Wen, H.X. Jiang, R.B. Qian, S.T. Liu, X.C. Tang, W.P. Huang, H.B. Chen, Micro-aerated fermentation enhances acetate production from high-rate activated sludge to supply carbon source for heterotrophic denitratation, Chem. Eng. J. 446 (2022) 18.
[53] H.Y. Wu, J.Y. Gao, D.H. Yang, Q. Zhou, W. Liu, Alkaline fermentation of primary sludge for short-chain fatty acids accumulation and mechanism, Chem. Eng. J. 160 (1) (2010) 1–7.
[54] Z.Y. Wu, D. Nguyen, T.Y.C. Lam, H.C. Zhuang, S. Shrestha, L. Raskin, S.K. Khanal, P.H. Lee, Synergistic association between cytochrome bd-encoded Proteiniphilum and reactive oxygen species (ROS)-scavenging methanogens in microaerobic-anaerobic digestion of lignocellulosic biomass, Water Res. 190 (2021) 11.
[55] W.M. Xie, B.J. Ni, T. Seviour, G.P. Sheng, H.Q. Yu, Characterization of autotrophic and heterotrophic soluble microbial product (SMP) fractions from activated sludge, Water Res. 46 (19) (2012) 6210–6217.
[56] Y. Xu, L.K. Zheng, H. Geng, R. Liu, X.H. Dai, Enhancing acidogenic fermentation of waste activated sludge via isoelectric-point pretreatment: Insights from physical structure and interfacial thermodynamics, Water Res. 185 (2020) 11.
[57] B. Yin, H.B. Liu, Y.Y. Wang, J. Bai, H. Liu, B. Fu, Improving volatile fatty acids production by exploiting the residual substrates in post-fermented sludge: Protease catalysis of refractory protein, Bioresour. Technol. 203 (2016) 124–131.
[58] N.J.W. Yu, B. Guo, Y.D. Zhang, L. Zhang, Y. Zhou, Y. Liu, Different micro-aeration rates facilitate production of different end-products from source-diverted blackwater, Water Res. 177 (2020) 8.
[59] Q. Yuan, R. Sparling, J.A. Oleszkiewicz, Waste activated sludge fermentation: Effect of solids retention time and biomass concentration, Water Res. 43 (20) (2009) 5180–5186.
[60] Z.W. Zhang, Y. Yu, H.B. Xi, Y.X. Zhou, Review of micro-aeration hydrolysis acidification for the pretreatment of toxic and refractory organic wastewater, J. Clean Prod. 317 (2021) 16.
[61] H.W. Zhao, D.S. Mavinic, W.K. Oldham, F.A. Koch, Controlling factors for simultaneous nitrification and denitrification in a two-stage intermittent aeration process treating domestic sewage, Water Res. 33 (4) (1999) 961–970.
[62] A.J. Zhou, W.Z. Liu, C. Varrone, Y.Z. Wang, A.J. Wang, X.P. Yue, Evaluation of surfactants on waste activated sludge fermentation by pyrosequencing analysis, Bioresour. Technol. 192 (2015) 835–840.
[63] P.J. Zhou, M.N.A. Meshref, B.R. Dhar, Optimization of thermal hydrolysis process for enhancing anaerobic digestion in a wastewater treatment plant with existing primary sludge fermentation, Bioresour. Technol. 321 (2021) 12.